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DATA SATURDAYS

Data Saturday Rheinland
2026-07-11

Who is burning your capacity

... and what can you do against it



Who am I

Or who makes that noise…

- Gregor Prohaska
- Operations Manager @ COSMO CONSULT
- „Some“ years of experience

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- Father of 2
- Curious by nature
- Don’t take myself to seriously ;-)

AGENDA

01 Understand

02 Monitor

03 Act & Protect

1

Understand

how fabric consumption works

Capacity, Capacity Sizes, Capacity Units and Billing Unit

Understanding

- Capacity – the car you drive (or set on fire)
  - Think of a VM – when it's running you pay for it
- Capacity Size – the fuel container
  - Defines the quantity of available fuel
- Capacity Units (CUs) – the fuel of fabric
  - Think of it as compute power per Second
- Billing Unit – package size of your fuel
  - calculated on 30-seconds increment

Capacity	CUs	Billing Unit	CUs/Day
F2	2	60	172,800
F4	4	120	345,600
F8	8	240	691,200
F16	16	480	1,382,400
F32	32	960	2,764,800
F64	64	1,920	5,529,600
…	…	…	…
F2048	2048	61,440	176,947,200



Capacity, Capacity Sizes, Capacity Units and Consumption

Understanding

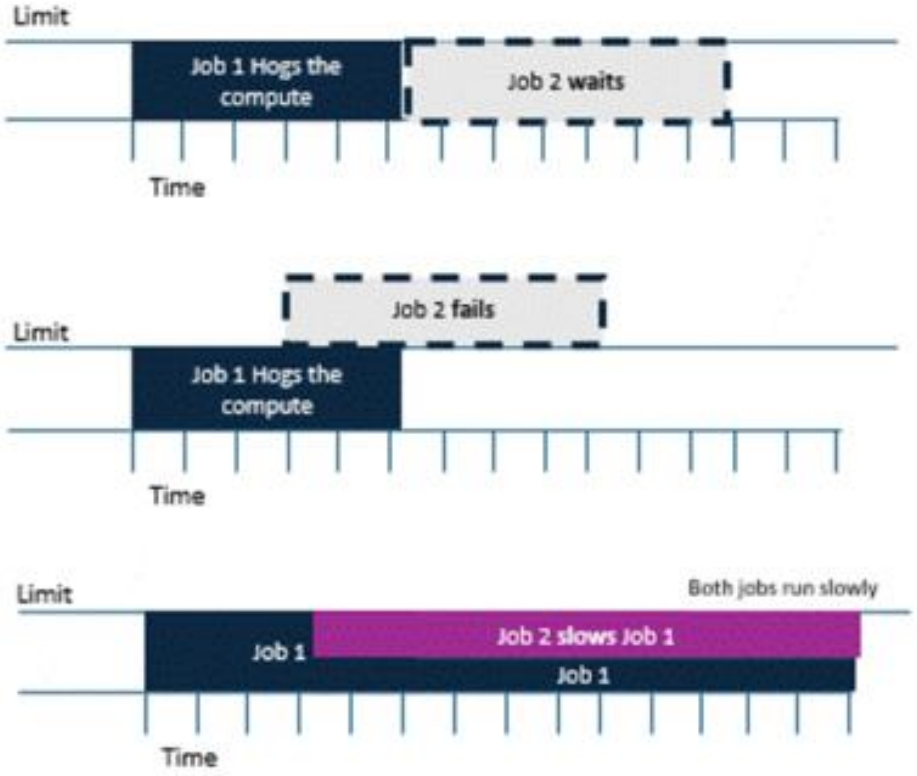
	Power BI						Direct Lake				Data Warehouse		Spark		
SKU	**Max memory (GB)**	**Max concurrent DirectQuery connections (per semantic model)**	**Max DirectQuery parallelism**	**Live connection (per second)**	**Max memory per query (GB)**	**Model refresh parallelism**	**Direct Lake rows per table (in millions)**	**Max Direct Lake model size on OneLake (GB)**	**Parquet files per table**	**Row groups per table**	**Baseline Capacity Units (CU)**	**Burstable Scale Factor**	**Spark VCores**	**Max Spark VCores with burst factor**	**Queue limit**
F2	3	5	1	2	1	1	300	10	1000	1000	2	1x - 32x	4	20	4
F4	3	5	1	2	1	2	300	10	1000	1000	4	1x - 16x	8	24	4
F8	3	10	1	3.75	1	5	300	10	1000	1000	8	1x - 12x	16	48	8
F16	5	10	1	7.5	2	10	300	20	1000	1000	16	1x - 12x	32	96	16
F32	10	10	1	15	5	20	300	40	1000	1000	32	1x - 12x	64	192	32
F64	25	50	4 -8	30	10	40	1500	Unlimited	5000	5000	64	1x - 12x	128	384	64
F128	50	75	6 -12	60	10	80	3000	Unlimited	5000	5000	128	1x - 12x	256	768	128
F256	100	100	8 - 16	120	10	160	6000	Unlimited	5000	5000	256	1x - 12x	512	1536	256
F512	200	200	10 - 20	240	20	320	12000	Unlimited	5000	5000	512	1x - 12x	1024	3072	512
F1024	400	200	12 -24	480	40	640	24000	Unlimited	10000	10000	1024	1x - 12x	2048	6144	1024
F2048	400	200	12 -24	960	40	1280	24000	Unlimited	10000	10000	2048	1x - 12x	4096	12288	2048

https://learn.microsoft.com/en-us/fabric/enterprise/powerbi/service-premium-what-is#capacities-and-skus
https://learn.microsoft.com/en-us/fabric/fundamentals/direct-lake-overview
https://learn.microsoft.com/en-us/fabric/data-warehouse/burstable-capacity
https://learn.microsoft.com/en-us/fabric/data-engineering/spark-job-concurrency-and-queueing



Bursting & Smoothing

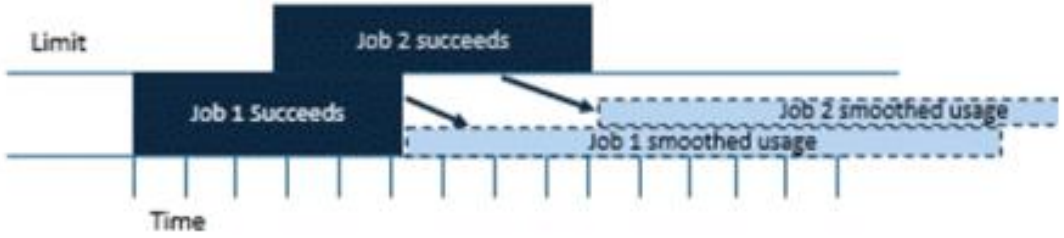
Understanding



Bursting lets both jobs **succeed**, running at **peak performance** to finish fast.



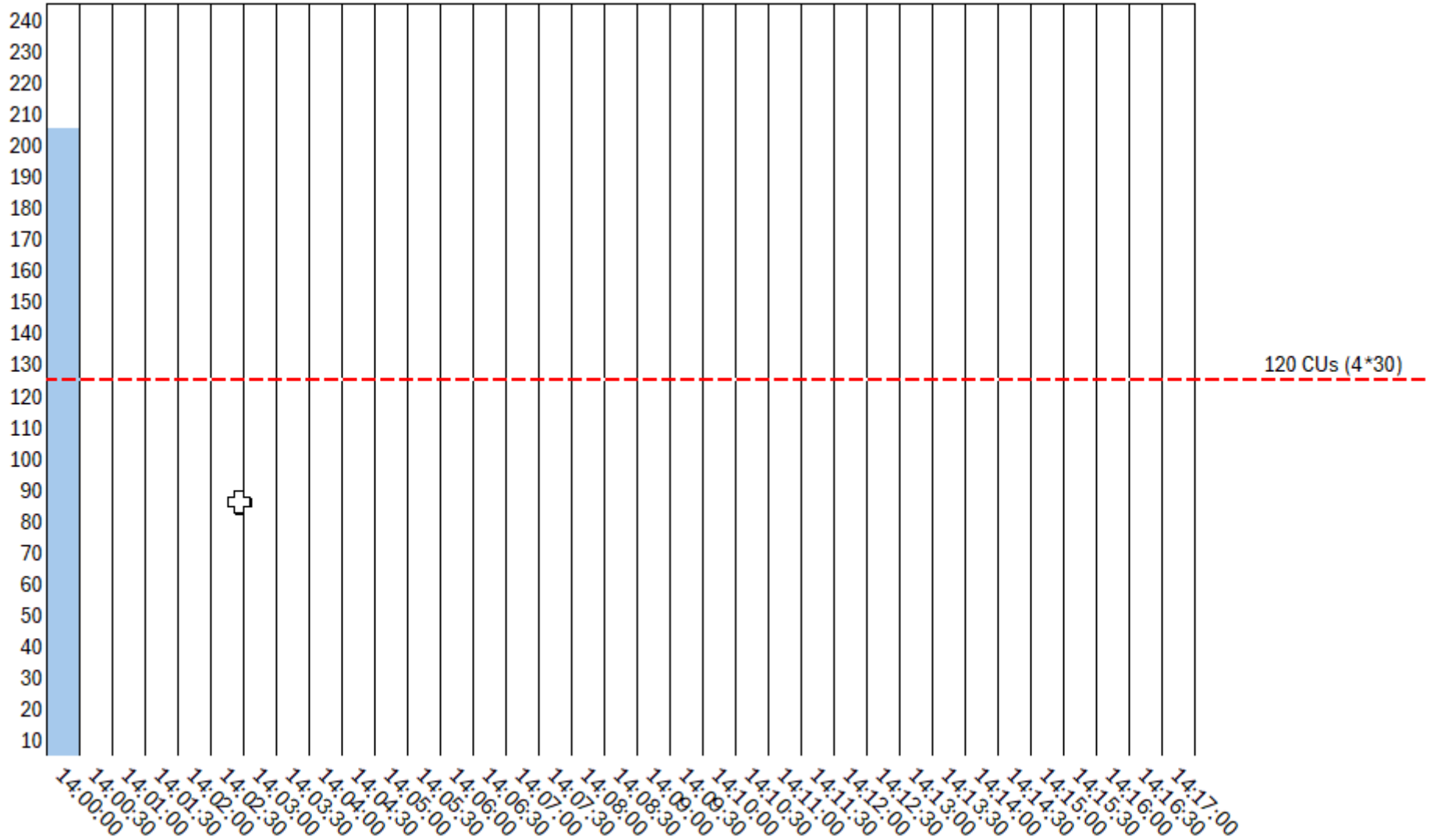
Smoothing distributes the **cost** of jobs over a longer time-period preventing scheduling issues.



https://learn.microsoft.com/en-us/fabric/enterprise/capacity-planning-plan-deployment

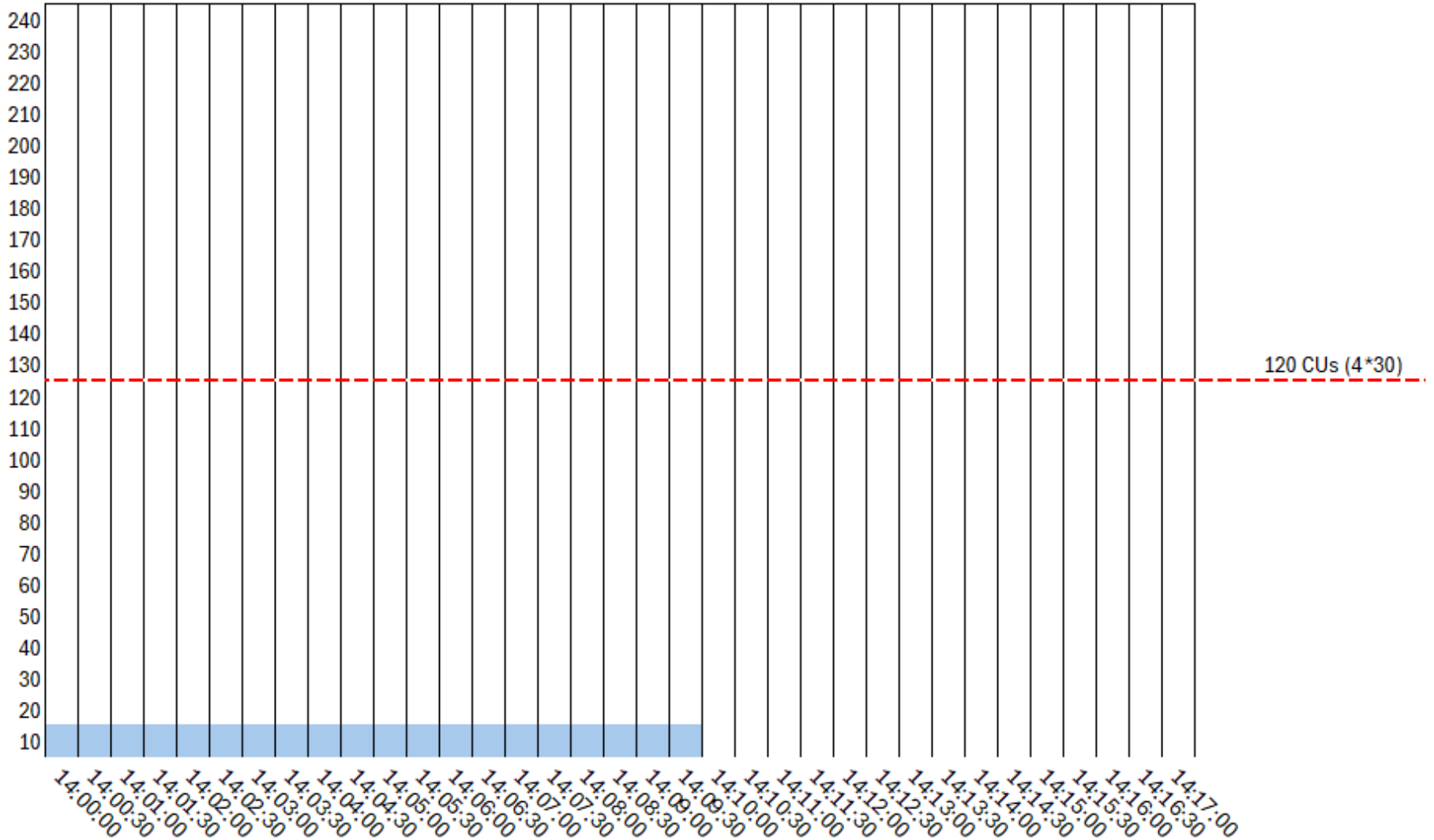
Bursting & Smoothing

Understanding



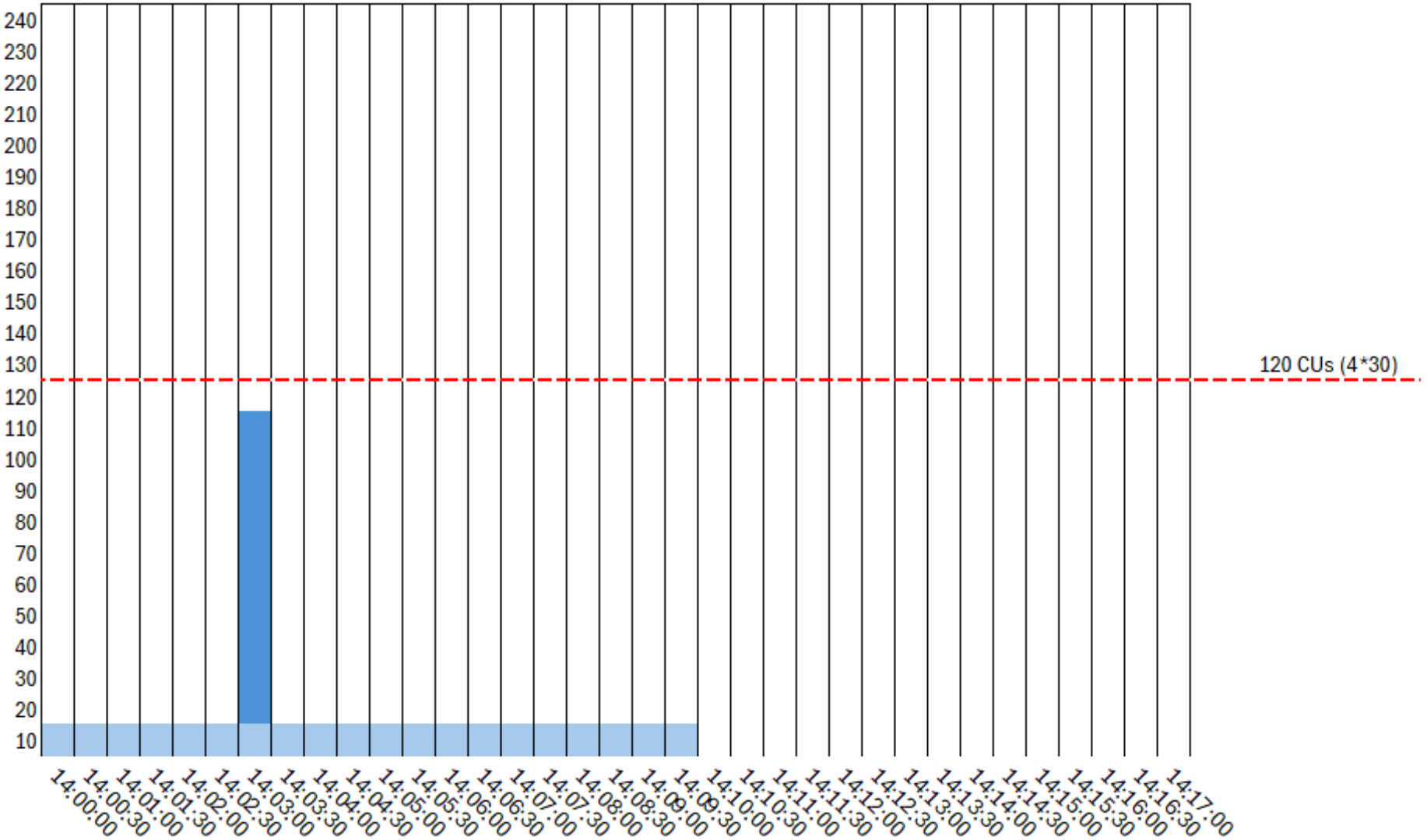
Bursting & Smoothing

Understanding



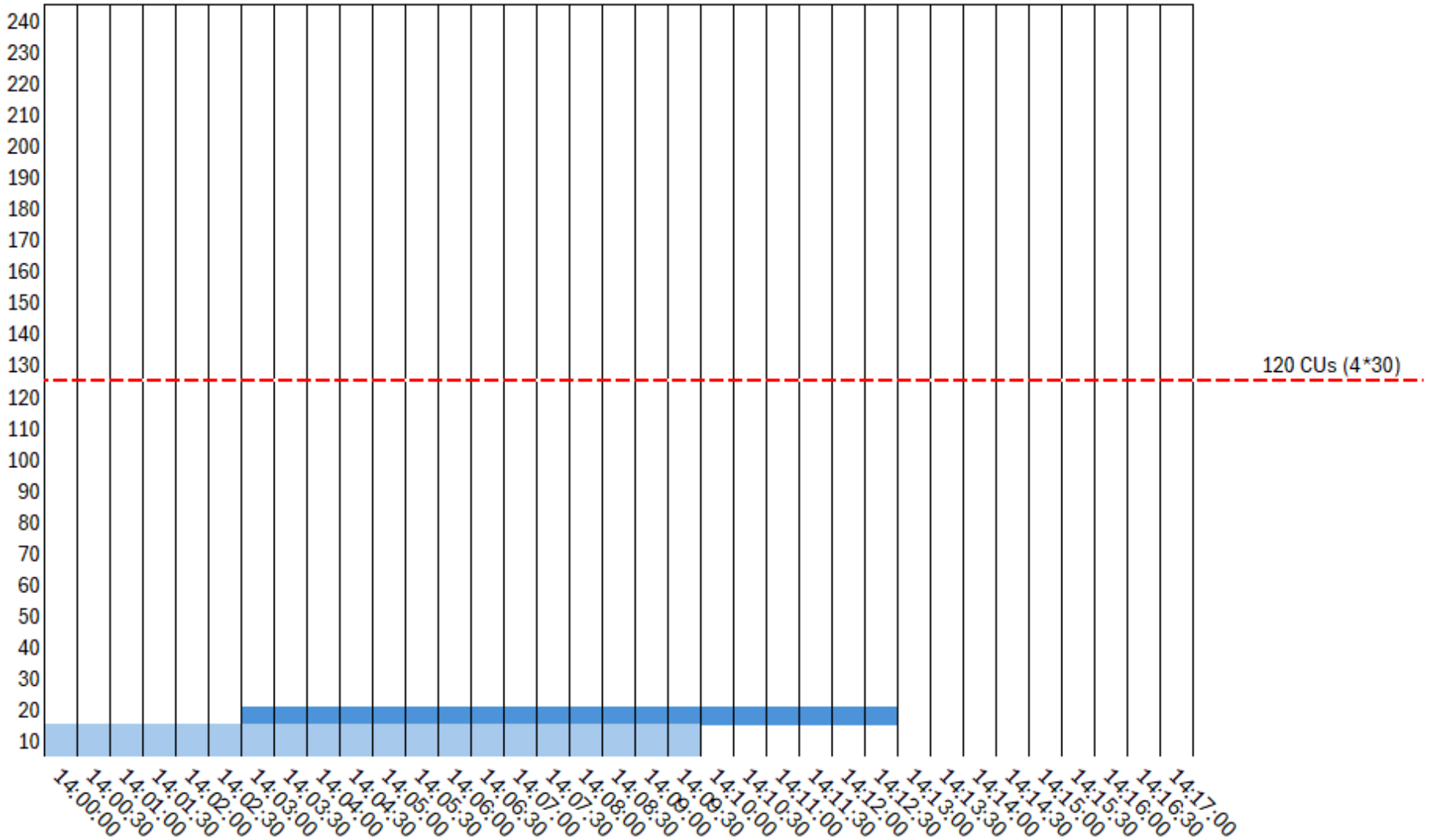
Bursting & Smoothing

Understanding



Bursting & Smoothing

Understanding



Capacity Throttling & Carry forward

Understanding

Future Smoothed Consumption – Policy Limits	Platform Policy	Experience Impact
Usage <= 10 minutes	Overage protection	Jobs can consume 10 minutes of future capacity use without throttling.
10 minutes < Usage <= 60 minutes	Interactive Delay	User requested interactive type jobs will be throttled.
60 minutes < Usage <= 24 hours	Interactive Rejection	User requested interactive type jobs will be rejected.
Usage > 24 hours	Background Rejection	User Scheduled background jobs will be rejected from execution.

$$\frac{CU\ Consumed\ for\ next\ X\ minutes + cumulative\ carry\ forward}{CU\ Budget\ for\ next\ X\ minutes} = Throttling\ Delay\ \%$$

https://blog.fabric.microsoft.com/en-US/blog/fabric-capacities-everything-you-need-to-know-about-whats-new-and-whats-coming/



Capacity Throttling & Carry forward

Understanding

Example:

- F64 Capacity → 64*30 = 1.920 CUs
- model refresh with 1 million CU → 1.000.000/2880 =347 CUs (rounded)
- Visual 1 refresh with 15.000 CU → 15.000/10 = 1.000 CUs
- Visual 2 refresh with 3.000 CU → 3.000/10 = 300 CUs

Current Time

	15:30:00	15:30:30	15:31:00	15:31:30	15:32:00	15:32:30	15:33:00	15:33:30	15:34:00	15:34:30	15:35:00	15:35:30	15:36:00	15:36:30	15:37:00	15:37:30
model refresh	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347
Visual 1 refresh		1500	1500	1500	1500	1500	1500	1500	1500	1500	1500					
Visual 2 refresh						300	300	300	300	300	300	300	300	300	300	
Total	347	1847	1847	1847	1847	2147	2147	2147	2147	2147	2147	647	647	647	647	347

https://www.linkedin.com/pulse/explaining-10-60-1440-minute-throttling-limits-part-3-matthew-farrow-q478e



Capacity Throttling & Carry forward

Understanding

Current Time

	15:32:00	15:32:30	15:33:00	15:33:30	15:34:00	15:34:30	15:35:00	15:35:30	15:36:00	15:36:30	15:37:00	15:37:30	15:38:00	15:38:30	15:39:00	15:39:30	15:40:00	15:40:30	15:41:00	15:41:30	Total
model refresh	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	347	
Visual 1 refresh	1500	1500	1500	1500	1500	1500															
Visual 2 refresh	300	300	300	300	300	300	300	300	300	300											
Total	2147	2147	2147	2147	2147	2147	647	647	647	647	347	347	347	347	347	347	347	347	347	347	
CU used in Window	1920	1920	1920	1920	1920	1920	647	647	647	647	347	347	347	347	347	347	347	347	347	347	17 578
Budget	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	1920	38 400
Carry Forward Add	227	227	227	227	227	227	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1 362

$$\frac{17578 + 1362}{38400} = 49{,}32\ \%\ \text{Interactive Delay (10 Min)}$$

$$\frac{CU\ Consumed\ for\ next\ X\ minutes + cumulative\ carry\ forward}{CU\ Budget\ for\ next\ X\ minutes}$$

$$= Throttling\ Delay\ \%$$

https://www.linkedin.com/pulse/explaining-10-60-1440-minute-throttling-limits-part-3-matthew-farrow-q478e

Pay-as-you-go & Reservation

Understanding

Pay as you go

- Can be stopped (paused)
- Additional costs may occur on stopping
- Can be scaled
- Items not accessible when stopped

Reservation

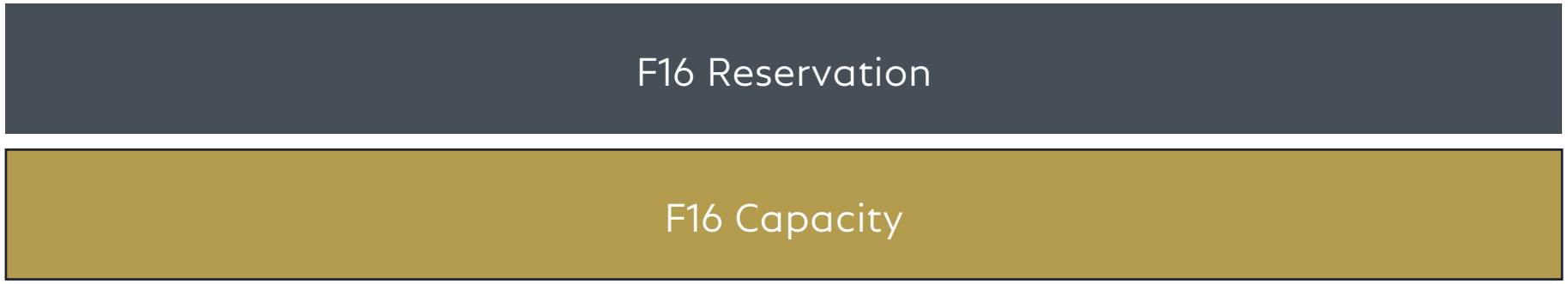
- Saves up to ~40%
- Commitment for 1 or 3 years
- Region-Bound & scoped
- Automatically applied



Pay-as-you-go & Reservation

Understanding

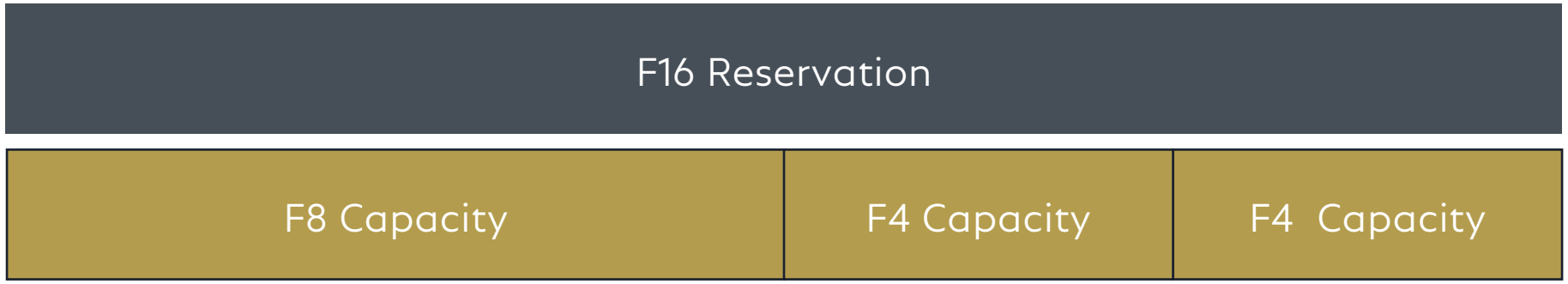
1 Capacity, covering reservation



Pay-as-you-go & Reservation

Understanding

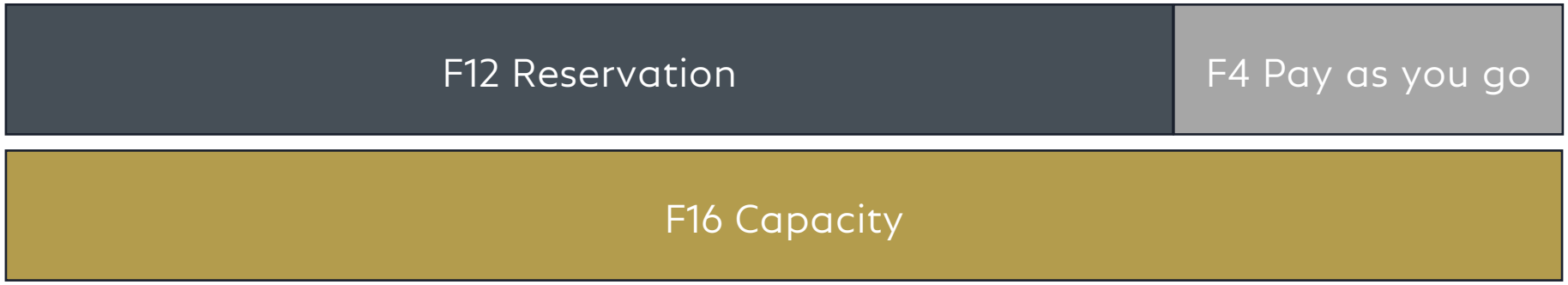
multiple Capacities, covering reservation



Pay-as-you-go & Reservation

Understanding

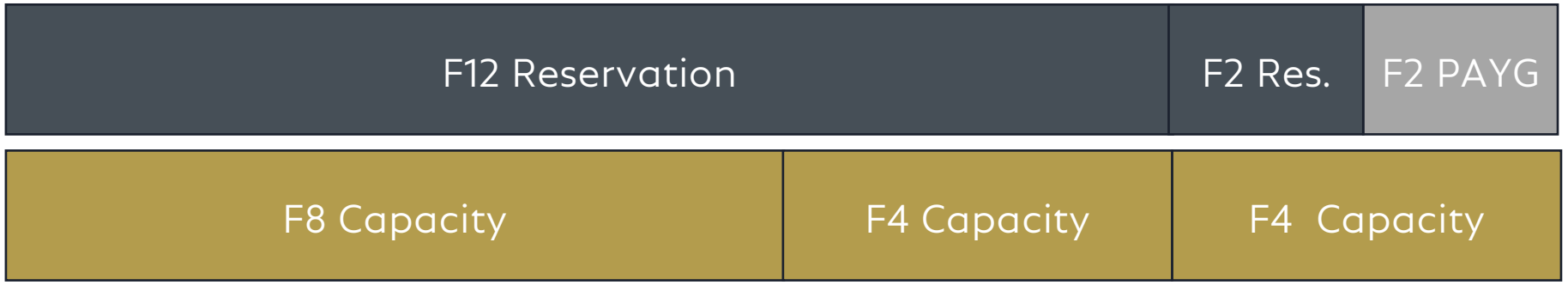
1 Capacity, not-covering reservation



Pay-as-you-go & Reservation

Understanding

multiple Capacities, multiple, not covering reservation



Summary

Understanding

- Capacity size **determines number of available CUs** and further workload specific factors.
- Bursting occurs when an operation **requires more CUs than available** during that period.
- Smoothing **always** takes place and distributes CU usage **evenly across future time points**.
- Throttling policies **become stricter** based **on** the **overload** factor.



2 Monitor

See what happens in your capacity



Tools

Monitoring

Built-in

- Monitor hub
- QueryInsights

Deployable

- Microsoft Fabric Chargeback Reporting (App)
- Microsoft Fabric Capacity Metrics (App)
- Fabric Platfrom Monitoring - Capacity Events



Summary

Monitoring

- Built-In solutions are **distributed across fabric** and **workload-specific**
- Capacity Metrics **slow** query performance, **data latency** up to 15 minutes but provides **in-depth information**
- Chargeback Reporting **focuses on originator** and is based on imported data
- Real-time solutions **consume CUs**, can be used for custom **notifications** (Activator) and are **customisable**



3

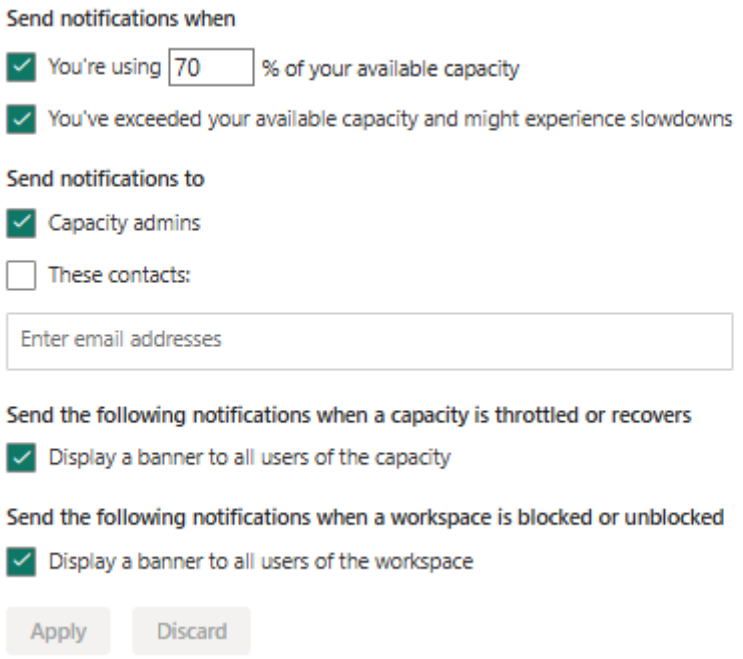
Act & Protect

What to do to protect your capacity

Setup Notifications & monitor regularly

Acting & Protecting

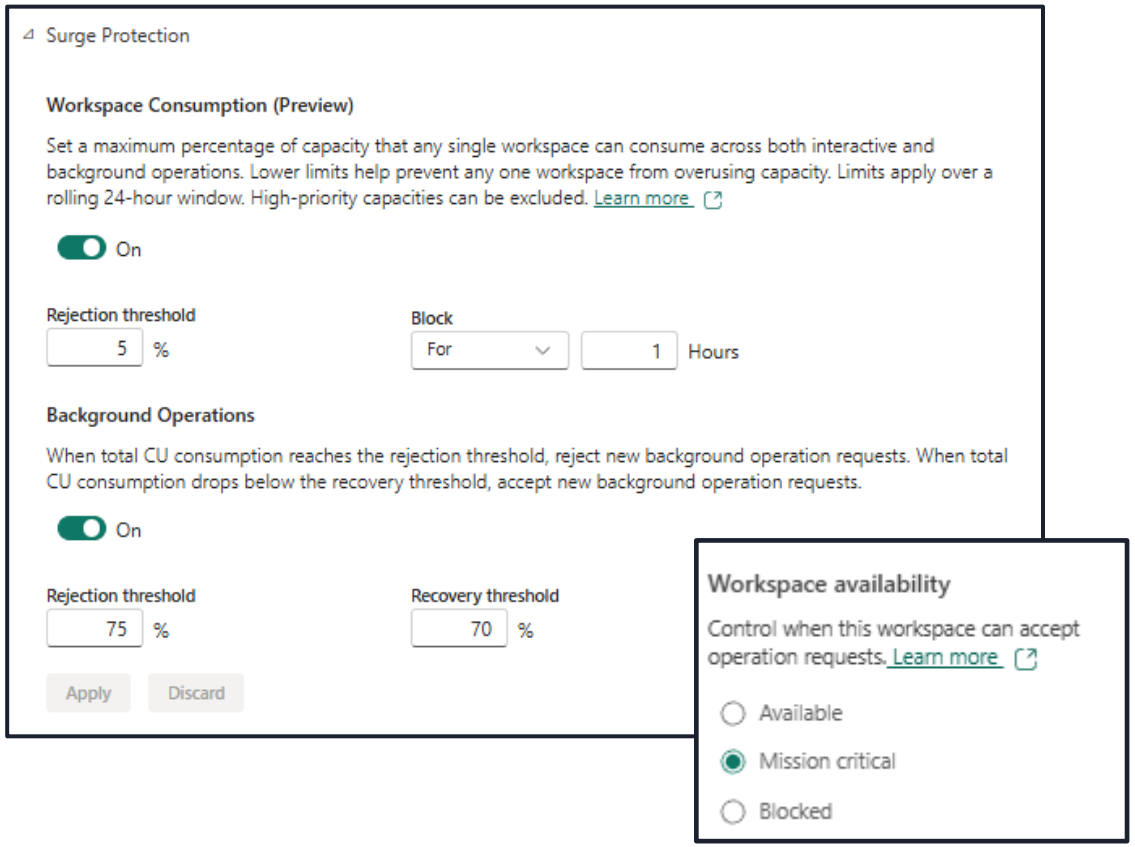
- Set-up a **warning before** anything happens
- Receive information **immediatley** if something happens
- Make the information **publicly available**



Surge Protection

Acting & Protecting

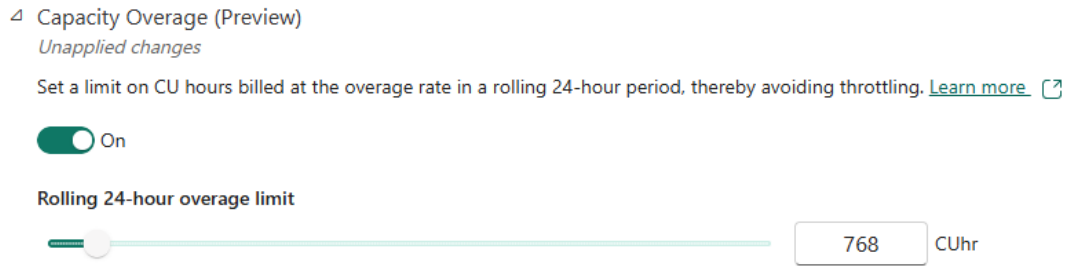
- Use Workspace Consumption to **block noisy neighbours** early
- Define your **mission critical** ones
- Enforce **early background rejection** if interactivity is more important



Capacity overage

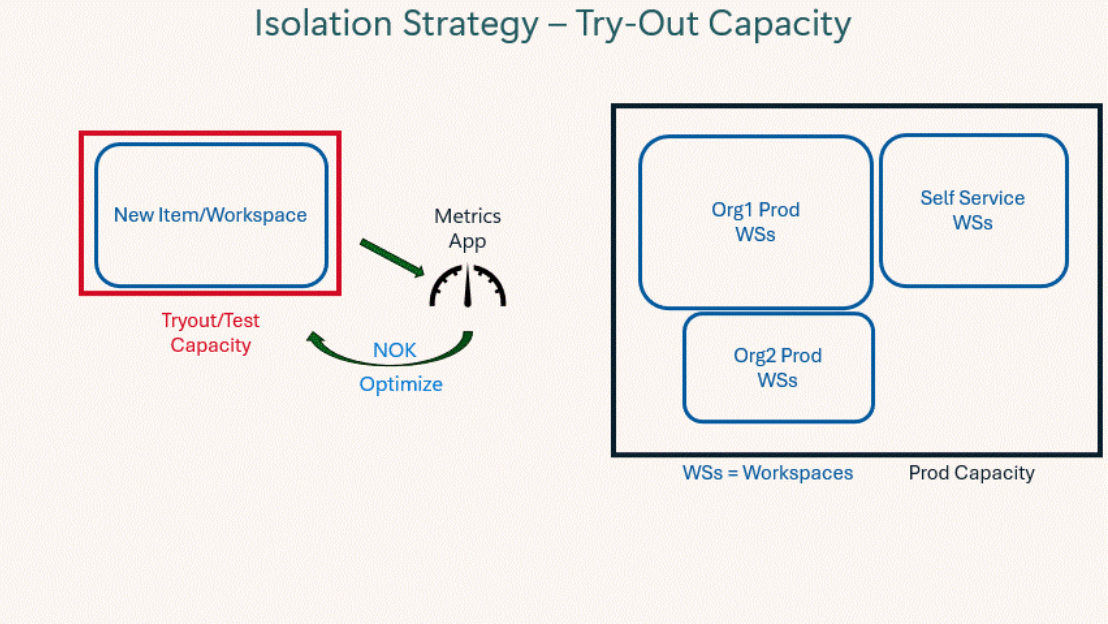
Acting & Protecting

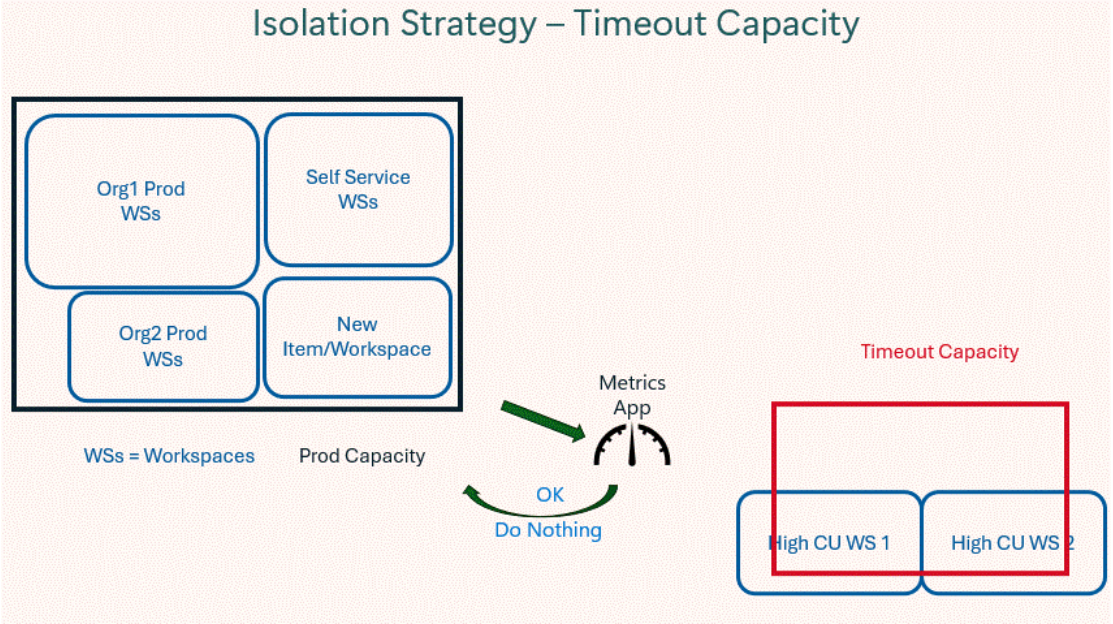
- Activate Capacity overage **if** capacity is **business critical**
- 3 x PAYG-Price



Capacity splitting

Acting & Protecting





https://learn.microsoft.com/en-us/fabric/enterprise/capacity-planning-scale-self-service-analytics

Summary

Acting & Protecting

- Use built-in notifiactions options – that way **you** and **your users stay informed**
- Have a **spare capacity readily available** (paused ;-))
- **find "the sweetspot"** for surge protection
- Use capacity overage **with caution** and for **closely monitored** workspaces (real-time)

Wrap Up

What I hope you have learned

- You **Understand** how Fabric **distributes** operational costs **over time**
- You know how to read **different tools** to **Monitor** your fabric capacity
- You are aware of different methods **to Protect** your capacity **and** how **to Act** if something happens



Thank you,

stay curious &
please give feedback



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Slides: https://bit.ly/DSRheinland2026